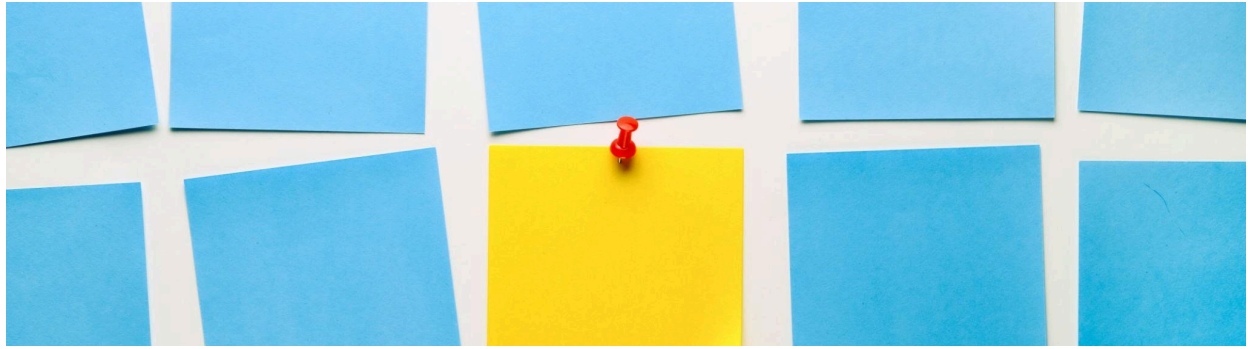




PHISHING EMAIL DETECTION & AWARENESS REPORT

1. Introduction

Phishing is one of the most common cyber attacks. Hackers send fake emails that look real. They trick people to click on bad links, download virus files, or share passwords and OTPs.

Most hacking does not happen because of weak computers, but because people don't know how to identify fake emails. This report will help normal users understand and avoid phishing.

Objectives

- To analyze real phishing email samples
- To find common signs of phishing
- To classify emails as Safe, Suspicious, or Phishing
- To explain attacks in very simple language
- To create awareness guidelines for employees

Tools Used

Email Header Analyzer: Google Message Header Tool,

MXToolbox - to check if sender is fake

URL Checker: VirusTotal, urlscan.io - to check links without clicking

Documentation: Google Docs

2. Email Sample Analyzed

Email-01: SAFE (Low Risk)

Subject: Your Internship Task 2 Assigned

From: Future Interns interns@futureinterns.com

Hi Kashish,

Welcome! Your Task 2 is live in the portal

Login here: futureinterns.com

For help contact: support@futureinterns.com

Thanks & Regards,

Future Interns Team

Analysis of Email-01:

When I opened this email, the first thing I checked was the sender. It says

interns@futureinterns.com which is the official domain of Future Interns. I went to their official website and it matches. No spelling trick like **future-interns** or

futureintern.com. Then I saw the greeting - it says "**Hi Kashish**" - my actual name. A

generic greeting can be a warning sign, but it is not sufficient by itself to classify an email as phishing. So this feels personal and real. There is **no urgency or fear** in this

mail. It just says Task 2 is live, login here. It doesn't ask for **OTP, password**, or say **"account will be locked in 24 hours"**. I also hovered on the login link, it showed futureinterns.com only. And when I checked on **VirusTotal**, it showed **0/89** - clean. **MXToolbox** shows **SPF** and **DKIM** pass. So after checking everything, I found **0 red flags**. **Verdict: SAFE** - This is a legit mail from Future Interns. No significant phishing indicators were identified during the checks performed. Users should still verify unexpected emails through official channels

Email-02: SUSPICIOUS (Medium Risk)

From: HR Team <hr-notice@company-docs-share.example>
Reply-To: hr-helpdesk@different-domain.example
Subject: Updated Leave Policy - Document Shared

Hello Employee,

An updated leave policy document has been shared with you.
Please review it via the link below:
<http://company-docs-share.example/policy>

No password is requested. If you have any doubt, please confirm with HR directly.

Thanks,
HR Team

Analysis of Email-02:

This one looked suspicious right from the start. The email comes from **hr-notice@company-docs-share.example** but if I click reply, it goes to **hr-helpdesk@different-domain.example** - a completely different domain. This is called **Reply-To mismatch**. In my internship company, HR always replies from the same domain. So why is the reply going somewhere else? That's the **biggest doubt**.

Second thing, it says **"Hello Employee"** not my actual name. Future Interns mail used **"Hi Kashish"** but this one is **generic**, like they sent the same mail to 100 people. The link is also company-docs-share.example - **not https**, and the domain company-docs-share.example is not our official company domain. It looks like a file sharing site trying to look like **internal HR**.

Good thing is it doesn't create fear like **"your account will be locked"** and doesn't directly ask for **password** or **OTP**. So it's not 100% phishing, but risky.

That's why I mark it as suspicious. I would not click the link. I will confirm with HR directly by calling them or opening a leave policy from the official portal.

Red Flags: 3 - Reply-To mismatch , Generic greeting , http link

Verdict: SUSPICIOUS (Medium Risk) - Verify first, don't click.

Email-03: PHISHING (High Risk) - CONFIRMED

From: support@msupdate.net

Subject: Microsoft account password change

Time: 4:09 PM (26 minutes ago)Microsoft account

Your password changedYour password for the Microsoft account ethan@hooksecurity.co was changed.If this was you, then you can safely ignore this email.Security info used:

Country/region: United States

Platform: iOS

Browser: Safari

IP address: 77.196.86.10If this wasn't you, your account has been compromised. Please follow these steps:Reset your password.Review your security info.Learn how to make your account more secure.You can also opt out or change where you receive security notifications.Thanks,
The Microsoft account team

Analysis of Email-03:

This email claims to be a security alert from Microsoft, but when I looked at the actual email address, it comes from support@msupdate.net. Real security notifications from Microsoft only come from official domains like microsoft.com or account.microsoft.com, not from third-party sites like msupdate.net.

To confirm this, I checked the technical details:

- **Email Headers:** Analyzing the header shows an **SPF Fail**, meaning the IP address that sent this email was never authorized by Microsoft to send messages on their behalf.
- **Return-Path:** The return path is set to bounce@msupdate.net, which does not match official Microsoft mail servers.
- **WHOIS Ownership:** I ran a WHOIS lookup on msupdate.net and found the registrant contact listed as support@phishingbox.com (owned by Ryan Hardesty), proving it has no connection to Microsoft Corporation.
- **VirusTotal:** Checking the domain on VirusTotal showed that 3 out of 89 security engines flagged it as malicious/phishing.

The email uses fear tactics ("your account has been compromised") to make the user panic and click "Reset your password" without thinking. Clicking that link would take the user to a fake login page designed to steal passwords.

- **Red Flags Identified:** 5 (Fake sender domain, SPF failure in headers, panic/fear tactic, no personalized name, flagged by WHOIS and VirusTotal)
- **Verdict: PHISHING (High Risk)** – Malicious email. Delete immediately and report to IT.

3. How The Attack Works

In Simple Words - What happens is, a hacker sends a mail that looks 100% like Microsoft. The user gets scared seeing "Your account has been compromised" and clicks Reset Password. That button doesn't go to the real Microsoft - it goes to a fake login page that the hacker made, which looks exactly like Microsoft login. When you type your password there, password goes to hacker, not to Microsoft. Then hackers can open your real Outlook, Teams, or Office account and steal data.

4. Prevention Guide

What I Learned

Do's:

- Always check the full email address, not just name. support@msupdate.net is not Microsoft.
- Before clicking any link, hover your mouse on it and see where it really goes.
- If the email says account hacked, don't click from mail - open Microsoft site directly by typing microsoft.com in browser.
- If not sure, ask the IT team or HR directly.

Don'ts:

- Don't click links that say "Reset in 24 hours or the account will be locked" - this is a fear trick.
- Don't share OTP or password via email reply.
- Don't trust mails that say "Hello Employee" or "Dear User".
- Be cautious with HTTP links, especially when an email asks you to enter sensitive information. Prefer verifying the website through the organization's official website. Real companies use https.

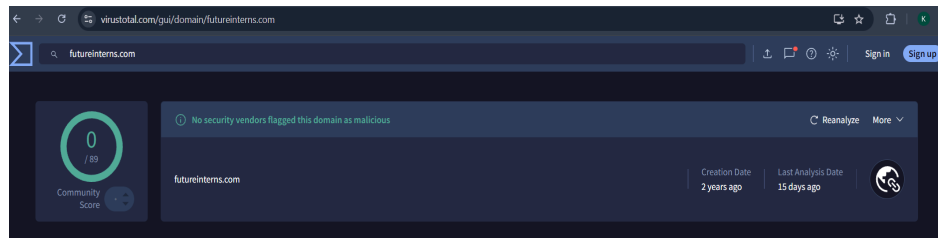
Email	Red Flags	Tools Result	Risk
-------	-----------	--------------	------

Email-01 Future Interns	0 Flags	SPF/DKIM PASS, VT 0/ 89 clean	SAFE- Low
Email-02 HR Policy	Reply - To mismatch, generic greeting , HTTP link	Domain mismatch in headers, unverified external link	SUSPICIOUS - Medium
Email-03 Microsoft	Fake domain msupdate.net , fear tactics	SPF Fail, VirusTotal 3/89 malicious, WHOIS shows PhishingBox	PHISHING - High

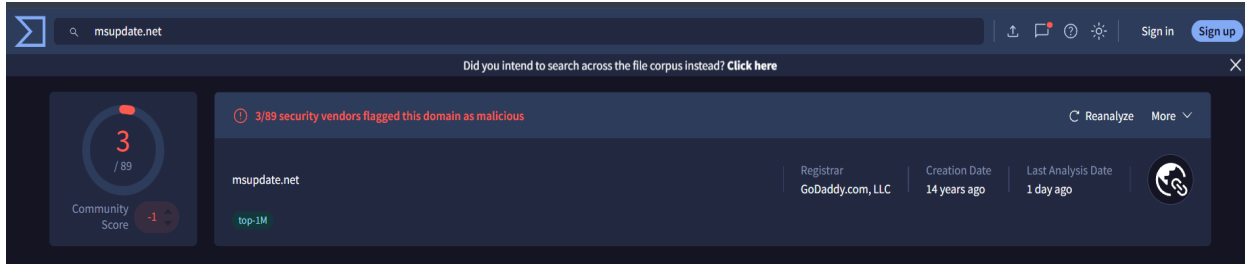
5.Conclusion

From this task I learned that phishing is not about high-level hacking - it's about tricking the human mind with fear and fake domains. By using simple tools like VirusTotal and Who.is and checking Reply-To mismatch, anyone can catch phishing mails. My 3 emails show a clear difference: SAFE has personal greeting and clean tools, SUSPICIOUS has reply-to mismatch, and PHISHING has fake domain and fear language with tool proof.

6.Section: Tool Evidence Screenshots



Screenshot 1 - VirusTotal 0/89 clean for futureinterns.com



Screenshot 2 - VirusTotal 3/89 malicious for msupdate.net

 Registrant Contact	
Name:	Ryan Hardesty
Street:	400 East Vine Street Suite 301
City:	Lexington
State:	KY
Postal Code:	40507
Country:	US
Phone:	+1.8594229165
Email:	support @phishingbox.com

Screenshot 3 - [Who.is](https://who.is) for msupdate.net showing support@phishingbox.com (not Microsoft)

